

Maryam Kameli

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[Google Scholar](#) | [LinkedIn](#) | [Website](#)

PhD candidate in computer science working on generative 3D representation and human visual perception. Current work builds photorealistic human avatars with 3D Gaussian Splatting and diffusion models, and measures how rendering choices change what people actually perceive. Four peer-reviewed publications, including IEEE TVCG.

EDUCATION

Doctor of Philosophy in **Computer Science**

University of Minnesota, Minneapolis, MN

September 2023 - Present

GPA: 3.71/4.00

Advisor: Prof. Victoria Interrante, Graphics and Visualization Lab.

Master of Science in **Computer Science**

University of Minnesota Duluth, Duluth, MN

January 2022 - May 2023

GPA: 3.85/4.00

Thesis: *Exploring Gender Differences in Brain Activity While Performing Different Tasks* [P3].

Bachelor of Science in **Electrical Engineering and Electronics**

K. N. Toosi University of Technology, Tehran, Iran

September 2017 - September 2021

GPA: 17.56/20.00

PUBLICATIONS

[P1] **M. Kameli**, K. Vaziri and V. Interrante. NPR Locomotion: Improving Spatial Awareness and Cybersickness with Sketch-Like Continuous Transitions. *IEEE Transactions on Visualization and Computer Graphics* (Proc. IEEE ISMAR), 2026.

[P2] **M. Kameli**, P. Bala and V. Interrante. Does Avatar Photorealism Enhance Social Presence? Comparing Captured vs. Crafted Models. *IEEE ISMAR Workshop on Hyperrealism in XR (Hype-XR)*, 2026.

[P3] **M. Kameli** and A. Khan. Unraveling Gender-Based Neural Variations in Task Performance: An EEG Study. *IEEE International Conference on E-health Networking, Application & Services (Healthcom)*, 2023.

[P4] **M. Kameli**, T. Zuraes and A. Khan. Improved Navigation for Social Robots Through Process Offloading. *IEEE International Conference on Computational Science and Computational Intelligence (CSCI)*, 2022.

RESEARCH EXPERIENCE

Photorealistic Avatar Synthesis with 3D Gaussian Splatting

January 2026 - Present

University of Minnesota, Minneapolis, MN

- Building a human avatar synthesis pipeline on 3D Gaussian Splatting with diffusion-based generative models, targeting real-time rendering at photorealistic fidelity
- Developing controllable fidelity levels inside one pipeline, so a single identity can be rendered along a photorealism gradient and evaluated for perceptual accessibility under sensory deficits
- Published perceptual results from this line comparing captured against crafted avatar models for social presence [P2]

Principal Research Assistant

June 2026 - August 2026

Hennepin Healthcare Research Institute (HHRI), Minneapolis, MN

- Designed a high-density EMG recording system mapping epidural spinal-cord-stimulation patterns to lower-limb muscle activation for a motor-complete spinal cord injury restoration study, with flexible electrode-array PCBs laid out in KiCad against spatial-sampling, impedance, and motion-artifact trade-offs drawn from primary literature
- Built a normalized SQLite schema and analysis pipeline for Gaussian-process Bayesian optimization of stimulation preferences, enabling cross-patient normalization and subgroup clustering across a 12-month clinical trial

Locomotion and Rendering Effects on Spatial Memory and Cybersickness

Sept 2024 - March 2026

University of Minnesota, Minneapolis, MN

- Designed sketch-like non-photorealistic continuous transitions for VR locomotion and showed they improve spatial awareness while reducing cybersickness; published in IEEE TVCG [P1]
- Built and deployed controlled remote experiments on Prolific end to end, from stimulus generation through data collection to analysis of ordinal, non-normal response data with Friedman and Wilcoxon signed-rank tests

Autonomous Mapping and Navigation for Social Robots

January 2022 - January 2023

University of Minnesota Duluth, Duluth, MN

- Integrated LiDAR and an NVIDIA Jetson Nano with a Pepper social robot over ROS, offloading perception and mapping to improve real-time reliability in human-occupied scenes [P4]
- Deployed the system across 9 nursing homes in Minnesota, and built an interactive web app for real-time facility mapping plus a customizable staff dashboard for resident activity sessions

Real-Time Human Fall Detection Framework

August 2022 - January 2023

University of Minnesota Duluth, Duluth, MN

- Built a real-time IoT backend with RESTful APIs streaming smartwatch time-series data to the cloud, plus a Python feature-engineering and classification pipeline separating falls from routine activity, containerized with Docker and served on Azure Kubernetes Service

Seizure Prediction with Deep Learning

August 2020 - September 2021

Undergraduate Thesis, K. N. Toosi University of Technology, Tehran, Iran

- Built CNN, LSTM, and Bi-LSTM models over EEG signals that predict epileptic seizures up to one hour before onset, turning raw recordings into clinically actionable early warnings

TECHNICAL SKILLS

Languages: C/C++, Python, SQL, Java, C#, JavaScript, HTML, CSS, R, Kotlin, Prolog, Assembly, VHDL

ML & Deep Learning: PyTorch, Hugging Face Transformers, TensorFlow, Scikit-learn, NumPy, Pandas, Bayesian optimization

Generative & 3D Vision: 3D Gaussian Splatting, diffusion models, Segment Anything (SAM), ViTPose, OpenCV, OpenGL

LLMs: Fine-tuning (BERT, GPT-2), retrieval-augmented generation, post-training, agentic research workflows

XR Development: Unity, Unreal Engine, Meta Quest, Apple Vision Pro

Systems & Tools: Linux, Docker, Kubernetes, Microsoft Azure, Git, ROS, PostgreSQL, MySQL, SQLite, KiCad, L^AT_EX

SELECTED PROJECTS

Markerless Human Motion Capture and Animation

Fall 2025

Computer Graphics

- Built an end-to-end video-to-animation pipeline with ViTPose for markerless pose estimation, Segment Anything for 3D mesh extraction, and OpenGL for rendering, using Savitzky-Golay filtering and spline interpolation to smooth noisy pose trajectories into realistic motion

Video-Based 3D Reconstruction and Segmentation

Fall 2023

Robot Vision and Computer Vision

- Implemented a pipeline reconstructing accurate 3D models from video, and combined SAM with monocular depth to improve segmentation under occlusion and transparency

Neural Machine Translation and LLM Fine-Tuning

Spring 2022

Advanced Natural Language Processing

- Built an attention-based transformer NMT system in PyTorch, using transfer learning to lift quality for low-resource languages, and fine-tuned BERT and GPT-2 on SuperGLUE to compare language understanding and text representation

TEACHING & MENTORING

Instructor, Introduction to Algorithms and Data Structures

Summer 2025

University of Minnesota

- Designed and delivered the full course including labs, projects, and rubrics, led a team of 5 teaching assistants, and earned evaluations above 5.5/6 on every measured dimension

Lead Teaching Assistant, Introduction to Artificial Intelligence

Fall 2024

University of Minnesota

- Developed weekly discussion plans across all sections, guided students through critical reading of research papers and technical writing, and advised the TA team on teaching strategy

Research Mentor

2022, 2026

REU Program, University of Minnesota; Master's Mentoring, University of Minnesota Duluth

- Mentored two REU researchers on VR/AR perception projects covering Unity, Meta Quest development, and Prolific study design, one with no prior coding experience, and guided an undergraduate through their first independent research project

SERVICE & OUTREACH

Reviewer | *ACM Symposium on Virtual Reality Software and Technology (VRST), Papers Track*

July 2026

Evaluated submitted VR/AR systems research for technical soundness, novelty, and clarity

Grant Reviewer | *Council of Graduate Students, University of Minnesota*

June 2026

Scored Conference, Career Development, and Research Support proposals under conflict-of-interest protocol

Volunteer Room Chair | *MinneAnalytics Data Tech Conference, Best Buy World Headquarters*

Summer 2026

Facilitated speaker sessions and moderated audience discussion

Teaching Assistant | *Machine Learning Summer Day Camp (K-12), University of Minnesota*

Summer 2024

Guided K-12 students through hands-on ML projects and a session with industry data scientists

Robotics Workshop Instructor | *University of Minnesota Duluth*

Spring 2022

Taught 40 chemistry students with no coding background, starting a recurring interdepartmental collaboration